

JEE MAIN REVISION MASTER: HYPERBOLA

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Asymptotes, Rectangular Hyperbola & Conjugate	3 PYQs	Hyperbola
Standard	Rectangular Hyperbola & Asymptotes	0 PYQs	Asymptotes & Geometry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Asymptotes, Rectangular Hyperbola & Conjugate	$e = \sqrt{1 + \frac{b^2}{a^2}}, \quad \text{LR: } y = \pm \frac{b}{a}x, \quad \text{Tangent: } y = mx \pm \sqrt{a^2 m^2 - b^2}, \quad \text{Rectangular: } xy = c^2 \implies e = \sqrt{2}$
Rectangular Hyperbola & Asymptotes	$\text{Asymptotes: } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0 \implies y = \pm \frac{b}{a}x, \quad \text{Rectangular: } a = b \iff e = \sqrt{2}, \quad xy = c^2 \implies (ct, c/t)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Asymptotes, Rectangular Hyperbola & Conjugate (Moderate Yield — 3 PYQs)

Core Principle: Focal distance difference property $|SP - S'P| = 2a$, eccentricity $e > 1$, asymptotes $y = \pm (b/a)x$, and rectangular hyperbola.

■ **Caution / JEE Pitfall:** For the conjugate hyperbola $-(x^2/a^2) + (y^2/b^2) = 1$, eccentricity is $e' = \sqrt{1 + a^2/b^2}$, and $(1/e^2) + (1/e'^2) = 1$.

■ **Speed Trick:** The eccentricity of ANY rectangular hyperbola ($a = b$ or $xy = c^2$) is always uniquely equal to $\sqrt{2}$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In the figure Statement-I: When $\alpha > \beta \geq 0$, the section is hyperbola Statement-II: When $\beta > 90^\circ$, the section is ellip..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "The distance between the foci of a hyperbola is 16 and its eccentricity is $\sqrt{2}$. Its equation is..." [Ans: N/A]
- [2018 Shift] Q#3 (MCQ): "The distance between the foci of a hyperbola is 16 and its eccentricity is $\sqrt{2}$. Its equation is..." [Ans: N/A]

3.2 Rectangular Hyperbola & Asymptotes (Standard Yield — 0 PYQs)

Core Principle: Equations and properties of asymptotes, angle between asymptotes $2 \tan^{-1}(b/a)$, and parametric properties of $xy = c^2$.

■ **Caution / JEE Pitfall:** The asymptotes of a hyperbola pass through its CENTER and are parallel to the conjugate hyperbola's asymptotes.

■ **Speed Trick:** Tangent to rectangular hyperbola $xy = c^2$ at $(ct, c/t)$ is $racxt + yt = 2c$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "In the figure Statement-I: When $\alpha > \beta \geq 0$, the section is hyperbola Statement-II: When $\beta > 90^\circ$, the section is ellip..." [Ans: N/A]
- **[2023 Shift] Q#2 (MCQ):** "The distance between the foci of a hyperbola is 16 and its eccentricity is $\sqrt{2}$. Its equation is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the figure Statement-I: When $\alpha > \beta \geq 0$, the section is hyperbola Statement-II: When $\beta > 90^\circ$, the section is ellipse Which of the following is correct?

- (A) Statement I is true, Statement II is false
- (B) Statement I is false, Statement II is true
- (C) Both the Statements are true
- (D) Both the Statements are false

Correct Answer: None

Solution Analysis: Let α be the semi-vertical angle of the cone, and let β be the angle between the cutting plane and the axis of the cone. The conic section depends on the comparison between α and β . If $0 \leq \beta < \alpha$, the plane cuts both parts, or nappes, of the cone. The section formed is a hyperbola. If $\beta = \alpha$, the plane is parallel to a generator of the cone. The section formed is a parabola. If $\alpha < \beta < 90^\circ$, the plane cuts only one nappe of the cone. The section formed is an ellipse. If $\beta = 90^\circ$, the plane is perpendicular to the axis of t